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Question Paper Code: 241MA004 QRN Code: 2410096 Regulation: 2023
B.E./B.Tech., DEGREE EXAMINATION, APRIL / MAY 2024
SECOND SEMESTER – B.E. ELECTRONICS AND COMMUNICATION
(ADVANCED COMMUNICATION TECHNOLOGY)
23MAI204 – LINEAR ALGEBRA AND COMPLEX VARIABLES
 Duration: Three hours Answer ALL questions Maximum: 100 marks

PART – A (10 x 2 = 20 marks) Answer all the Questions		CO No.	Revised Bloom's Cognitive Level
			Question
1.	Give two direct methods to solve a system of linear equations.	CO1	R
2.	Give any two properties of Permutation Matrix.	CO1	R
3.	Define a vector space.	CO2	R
4.	Determine by inspection if the given set is linearly dependent $[1,7,6],[2,0,9],[3,1,5],[4,1,8]$.	CO2	U
5.	Define row and column space.	CO3	R
6.	Define nullity.	CO3	R
7.	Define Conformal mapping.	CO4	R
8.	Find the map of the circle $ z =3$ under the transformation $w=2z$.	CO4	U
9.	Evaluate $\int_c \frac{dz}{2z-3}$ where c is the circle $ z =1$.	CO5	U
10.	State Cauchy's Residue theorem.	CO5	R

PART – B (1 x 16 =16 marks) (Scenario based – Compulsory Question)			Marks	CO No.	Revised Bloom's Cognitive Level
11.		Solve $x+2y+3z=10, 8x+y+2z=13, 2x+3y+z=13$ by LU Factorization.	16	CO1	Ap

PART – C (4 x 16 = 64 marks) (Answer any 4 out of 7 Questions)			Marks	CO No.	Revised Bloom's Cognitive Level
12.	(i)	Use Gauss - Jordan Method to find the inverse of $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & 5 \\ 1 & -1 & 0 \end{pmatrix}$.	8	CO1	Ap
	(ii)	Solve by Gauss Elimination Method $3x+4y+5z=18, 2x-y+8z=13, 5x-2y+7z=20$.	8	CO1	Ap
13.		Is the set of all pairs of real numbers with the operations $(x,y)+(x'+y')=(x+x'+1,y+y'+1)$ and $k(x,y)=(kx,ky)$ a vector space? Verify all the axioms and list the axioms that fail to hold if it is not a vector space.	16	CO2	Ap
14.	(i)	Find the spanning set for the null space of the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$.	8	CO3	Ap
	(ii)	Show that A is an invertible matrix and B is its inverse.	8	CO3	Ap
15.	(i)	If $f(z)$ is a regular function, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial y^2} \right) f(z) ^2 = 4 f'(z) ^2$	8	CO4	Ap
	(ii)	Find the bilinear transformation which maps the points $z=\infty, i, 0$ of the z -plane onto $w=0, i, \infty$ of the w -plane.	8	CO4	Ap

16.	(i)	Find the analytic function whose imaginary part is $e^x (x \sin y + y \cos y)$.	8	CO4	Ap
	(ii)	Find the image of the circle $ z - 1 = 1$ in the complex plane under the mapping $w = \frac{1}{z}$.	8	CO4	Ap
17.	(i)	Evaluate $\int_C \frac{z-1}{(z+1)(z-2)} dz$, where C is the circle $ z - i = 2$ using Cauchy's residue theorem.	8	CO5	Ap
	(ii)	Expand $f(z) = \frac{z^2-1}{(z+3)(z+2)}$ in a Laurent's series if (i) $ z > 3$ (ii) $2 < z < 3$	8	CO5	Ap
18.	(i)	Evaluate $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ where c is the circle $ z =3$	8	CO5	Ap
	(ii)	Evaluate $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)(x^2+b^2)} dx, a > 0, b > 0$	8	CO5	Ap